

IT-Simplera Solutions



Week 7 Task Report

Enterprise VPN, Firewall, and Secure Access Implementation

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Date Of Submission

15 August 2026

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1. Assignment Overview

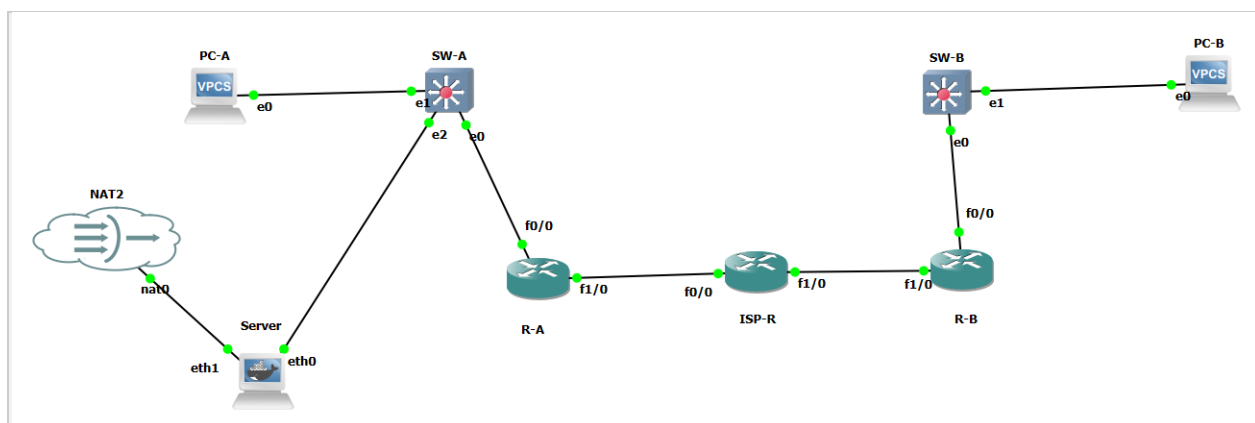
This assignment required the design, configuration, and verification of a secure enterprise network for IT-Simplera Solutions, connecting two branch offices (Branch A and Branch B) through a simulated internet cloud in GNS3. The network implements an IPsec Site-to-Site VPN between branches, a Zone-Based Firewall (ZBF) protecting each branch LAN, AAA authentication with SSH for secure device management, OSPF with MD5 authentication for secure routing, and centralized monitoring via Syslog, NTP, and SNMPv3.

1.1 Objectives

- Implement secure branch-to-branch communication using IPsec Site-to-Site VPN technology.
- Configure and verify Zone-Based Firewall policies to protect enterprise resources.
- Deploy AAA authentication and secure remote management using SSH.
- Implement device hardening and secure routing practices to enhance enterprise security.
- Monitor, verify, and troubleshoot security configurations using Syslog, NTP, and SNMP.

2. Network Topology Design

The topology consists of 5 core devices: two branch edge routers (R-A, R-B), a middle router (ISP-R) simulating the public internet between the branches, and two Layer 2 switches (SW-A, SW-B) forming each branch LAN. A Docker-based Alpine Linux server acts as the centralized Syslog, NTP, and SNMP monitoring host, attached to the Branch A LAN. R-A and R-B each run a Zone-Based Firewall separating their internal LAN (zone INSIDE) from the WAN-facing interface (zone OUTSIDE), and an IPsec tunnel is built directly between their WAN interfaces, riding across ISP-R which has no knowledge of the internal branch subnets — exactly as it would over the real internet.



#	Device	Role
1	R-A	Branch A edge router — LAN gateway, ZBF, IPsec peer, OSPF
2	ISP-R	Simulated internet — routes WAN traffic only, no knowledge of LAN subnets
3	R-B	Branch B edge router — LAN gateway, ZBF, IPsec peer, OSPF
4	SW-A	Branch A Layer 2 switch
5	SW-B	Branch B Layer 2 switch
6	Server (Alpine, Docker)	Centralized Syslog, NTP, SNMPv3 monitoring host
7	PC-A / PC-B	End-user hosts (VPCS) used for end-to-end connectivity testing

3. IP Addressing Plan

Device	Interface	IP Address
R-A	FastEthernet0/0 (LAN)	192.168.10.1 /24
R-A	FastEthernet1/0 (WAN)	203.0.113.1 /30
ISP-R	FastEthernet0/0	203.0.113.2 /30
ISP-R	FastEthernet1/0	203.0.113.5 /30
R-B	FastEthernet1/0 (WAN)	203.0.113.6 /30
R-B	FastEthernet0/0 (LAN)	192.168.20.1 /24
SW-A	Vlan1 (management)	192.168.10.2 /24
SW-B	Vlan1 (management)	192.168.20.2 /24
PC-A	eth0	192.168.10.10 /24

Device	Interface	IP Address
PC-B	eth0	192.168.20.10 /24
Server	eth0	192.168.10.100 /24

VPN interesting traffic (ACL 101): permits 192.168.10.0/24 ↔ 192.168.20.0/24, matched on both R-A and R-B with source/destination reversed accordingly.

4. Security Architecture

Defense is applied in layers on each branch edge router:

- Perimeter routing security — OSPF adjacencies between each branch router and ISP-R are authenticated with MD5 (message-digest-key), preventing rogue routers from injecting false routes. The LAN-facing interface is passive, so no OSPF hellos are sent toward end users.
- Confidentiality in transit — all traffic between the two branch LANs crosses an IPsec Site-to-Site VPN (ISAKMP/IKE Phase 1 with AES-256/SHA-256/DH group 14, IPsec Phase 2 with AES-256/SHA-256-HMAC in tunnel mode). ISP-R never sees branch-internal addressing, only the encrypted ESP payload between the two public WAN endpoints.
- Perimeter access control — a Zone-Based Firewall on each router places the LAN in zone INSIDE and the WAN in zone OUTSIDE. A stateful inspect policy allows LAN-initiated TCP/UDP/ICMP sessions out and their return traffic, while unsolicited inbound sessions from OUTSIDE are dropped by the implicit class-default drop.
- Secure management — SSHv2 replaces Telnet for remote administration, backed by AAA authentication against a local user database; console and VTY access are both gated through AAA login.
- Centralized visibility — Syslog, NTP, and SNMPv3 (authPriv) point to a single monitoring host, giving synchronized timestamps, tamper-resistant polling, and a central audit trail for both branches.

5. Device Configurations

The complete configuration applied to each device is listed below. All FastEthernet WAN links use OSPF MD5 authentication; the IPsec tunnel is built directly between R-A and R-B's WAN addresses; the Zone-Based Firewall is applied identically (mirrored) on both branch routers. AAA, SSH, and hardening commands (Section 5.6) are applied last, after core connectivity and security services were verified independently.

5.1 R-A

```
enable
terminal length 0
configure terminal
hostname R-A
no ip domain-lookup
ip domain name itsimplera.local
interface FastEthernet0/0
  description LAN-BranchA
  ip address 192.168.10.1 255.255.255.0
  no shutdown
  exit
interface FastEthernet1/0
  description WAN-to-ISP
  ip address 203.0.113.1 255.255.255.252
  no shutdown
  exit
router ospf 1
  network 192.168.10.0 0.0.0.255 area 0
  network 203.0.113.0 0.0.0.3 area 0
  passive-interface FastEthernet0/0
  exit
interface FastEthernet1/0
  ip ospf message-digest-key 1 md5 CiscoOSPF123
  ip ospf authentication message-digest
  exit
crypto isakmp policy 10
  encryption aes 256
  hash sha256
  authentication pre-share
  group 14
  lifetime 86400
  exit
crypto isakmp key VpnKey123! address 203.0.113.6
access-list 101 permit ip 192.168.10.0 0.0.0.255
192.168.20.0 0.0.0.255
crypto ipsec transform-set TSET esp-aes 256 esp-sha256-hmac
  mode tunnel
  exit
```

```
crypto map VPNMAP 10 ipsec-isakmp
  set peer 203.0.113.6
  set transform-set TSET
  match address 101
  exit
interface FastEthernet1/0
  crypto map VPNMAP
  exit
zone security INSIDE
exit
zone security OUTSIDE
exit
class-map type inspect match-any INSIDE-TRAFFIC
  match protocol tcp
  match protocol udp
  match protocol icmp
  exit
policy-map type inspect INSIDE-TO-OUTSIDE
  class type inspect INSIDE-TRAFFIC
    inspect
  exit
exit
zone-pair security IN-OUT source INSIDE destination OUTSIDE
  service-policy type inspect INSIDE-TO-OUTSIDE
  exit
interface FastEthernet0/0
  zone-member security INSIDE
  exit
interface FastEthernet1/0
  zone-member security OUTSIDE
  exit
no ip http server
no ip http secure-server
clock timezone UTC 0
logging host 192.168.10.100
logging trap informational
logging source-interface FastEthernet0/0
ntp server 192.168.10.100
snmp-server group SNMPGROUP v3 priv
```

```
snmp-server user SNMPUSER SNMPGROUP v3 auth sha AuthPass123
priv aes 128 PrivPass123
snmp-server host 192.168.10.100 version 3 priv SNMPUSER
end
write memory
```

5.2 ISP-R

```
enable
terminal length 0
configure terminal
hostname ISP-R
no ip domain-lookup
interface FastEthernet0/0
  description to-R-A
  ip address 203.0.113.2 255.255.255.252
  no shutdown
  exit
interface FastEthernet1/0
  description to-R-B
  ip address 203.0.113.5 255.255.255.252
  no shutdown
  exit
router ospf 1
  network 203.0.113.0 0.0.0.3 area 0
  network 203.0.113.4 0.0.0.3 area 0
  exit
interface FastEthernet0/0
  ip ospf message-digest-key 1 md5 CiscoOSPF123
  ip ospf authentication message-digest
  exit
interface FastEthernet1/0
  ip ospf message-digest-key 1 md5 CiscoOSPF123
  ip ospf authentication message-digest
  exit
end
write memory
```

5.3 R-B


```
enable
terminal length 0
configure terminal
hostname R-B
no ip domain-lookup
ip domain name itsimplera.local
interface FastEthernet0/0
  description LAN-BranchB
  ip address 192.168.20.1 255.255.255.0
  no shutdown
  exit
interface FastEthernet1/0
  description WAN-to-ISP
  ip address 203.0.113.6 255.255.255.252
  no shutdown
  exit
router ospf 1
  network 192.168.20.0 0.0.0.255 area 0
  network 203.0.113.4 0.0.0.3 area 0
  passive-interface FastEthernet0/0
  exit
interface FastEthernet1/0
  ip ospf message-digest-key 1 md5 CiscoOSPF123
  ip ospf authentication message-digest
  exit
crypto isakmp policy 10
  encryption aes 256
  hash sha256
  authentication pre-share
  group 14
  lifetime 86400
  exit
crypto isakmp key VpnKey123! address 203.0.113.1
access-list 101 permit ip 192.168.20.0 0.0.0.255
192.168.10.0 0.0.0.255
crypto ipsec transform-set TSET esp-aes 256 esp-sha256-hmac
  mode tunnel
  exit
crypto map VPNMAP 10 ipsec-isakmp
```

```
set peer 203.0.113.1
set transform-set TSET
match address 101
exit
interface FastEthernet1/0
crypto map VPNMAP
exit
zone security INSIDE
exit
zone security OUTSIDE
exit
class-map type inspect match-any INSIDE-TRAFFIC
match protocol tcp
match protocol udp
match protocol icmp
exit
policy-map type inspect INSIDE-TO-OUTSIDE
class type inspect INSIDE-TRAFFIC
inspect
exit
exit
zone-pair security IN-OUT source INSIDE destination OUTSIDE
service-policy type inspect INSIDE-TO-OUTSIDE
exit
interface FastEthernet0/0
zone-member security INSIDE
exit
interface FastEthernet1/0
zone-member security OUTSIDE
exit
no ip http server
no ip http secure-server
clock timezone UTC 0
logging host 192.168.10.100
logging trap informational
logging source-interface FastEthernet0/0
ntp server 192.168.10.100
snmp-server group SNMPGROUP v3 priv
snmp-server user SNMPUSER SNMPGROUP v3 auth sha AuthPass123
```

```
priv aes 128 PrivPass123
snmp-server host 192.168.10.100 version 3 priv SNMPUSER
end
write memory
```

5.4 SW-A / SW-B

```
enable
terminal length 0
configure terminal
hostname SW-A
no ip domain-lookup
interface range gigabitEthernet 0/0-3
    switchport mode access
    switchport access vlan 1
    no shutdown
exit
interface vlan 1
    ip address 192.168.10.2 255.255.255.0
    no shutdown
exit
ip default-gateway 192.168.10.1
end
write memory

! SW-B identical, using 192.168.20.2 / gateway 192.168.20.1
```

5.5 PC-A / PC-B (VPCS)

```
ip 192.168.10.10/24 192.168.10.1      ! PC-A
ip 192.168.20.10/24 192.168.20.1    ! PC-B
```

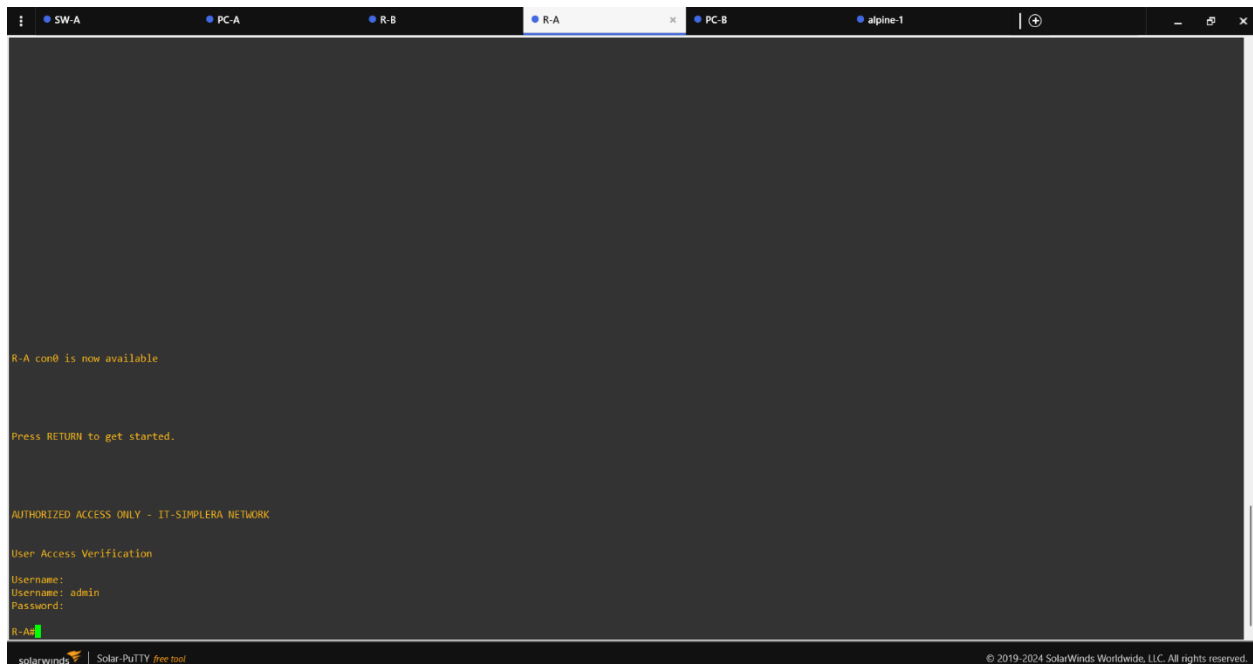
5.6 Monitoring Server (Alpine, Docker)

```
ip addr add 192.168.10.100/24 dev eth0
ip link set eth0 up
ip route add default via 192.168.10.1
echo "nameserver 8.8.8.8" > /etc/resolv.conf
apk update
apk add rsyslog chrony net-snmp net-snmp-tools
```

5.7 AAA, SSH, Hardening (applied on R-A and R-B)

Applied after core VPN/firewall/OSPF connectivity was verified, so that any lockout during testing would not block troubleshooting of the security services themselves.

```
username admin privilege 15 secret Admin@123
aaa new-model
aaa authentication login default local
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
  transport input ssh
  login authentication default
exit
line con 0
  login authentication default
  exec-timeout 5 0
exit
service password-encryption
enable secret Enable@123
banner motd #
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#
```



6. Verification Plan

6.1 OSPF Adjacency

```
show ip ospf neighbor
```

Result obtained on R-A:

Neighbor ID	Pri	State	Dead Time	Address
Interface				
203.0.113.5	1	FULL/BDR	00:00:36	
203.0.113.2		FastEthernet1/0		

OSPF adjacency reached FULL between R-A and ISP-R, confirmed authenticated via MD5 (no adjacency forms if the key mismatches).

6.2 IPsec VPN — ISAKMP (Phase 1)

```
show crypto isakmp sa
```

IPv4 Crypto	ISAKMP SA		
dst	src	state	conn-id
status			
203.0.113.6	203.0.113.1	QM_IDLE	1001
ACTIVE			

QM_IDLE / ACTIVE confirms Phase 1 (ISAKMP/IKE) negotiation completed successfully between the two WAN endpoints.

```
SW-A PC-A R-B R-A PC-B
R-A#
R-A#enable
R-A#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R-A(config)#policy-map type inspect OUTSIDE-TO-INSIDE
R-A(config-pmap)# class type inspect INSIDE-TRAFFIC
R-A(config-pmap-c)# inspect
R-A(config-pmap-c)# exit
R-A(config-pmap)#zone-pair security OUT-IN source OUTSIDE destination INSIDE
R-A(config-sec-zone-pair)# service-policy type inspect OUTSIDE-TO-INSIDE
R-A(config-sec-zone-pair)# exit
R-A(config)#end
R-A#write memory
*Aug 14 23:14:06.915: %SYS-5-CONFIG_I: Configured from console by console
R-A#write memory
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R-A#sh
R-A#show ospf ne
R-A#show ospf neighbor
R-A#show ospf neighbor
R-A#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
203.0.113.5    1     FULL/DR         00:00:36    203.0.113.2    FastEthernet1/0
R-A#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst            src            state          conn-id status
203.0.113.6    203.0.113.1    QM_IDLE        1001 ACTIVE

IPv6 Crypto ISAKMP SA
R-A#
```

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6.3 IPsec VPN — IPsec SAs (Phase 2)

```
show crypto ipsec sa

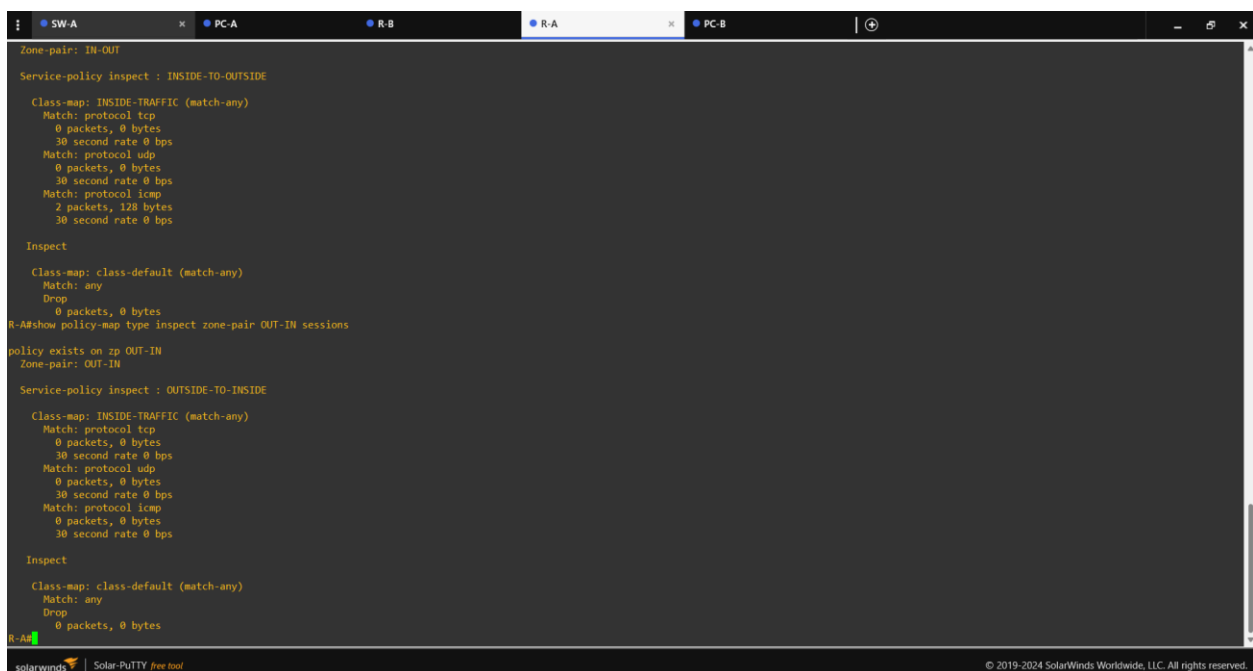
local ident (addr/mask/prot/port):
(192.168.10.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port):
(192.168.20.0/255.255.255.0/0/0)
current_peer 203.0.113.6 port 500
    #pkts encaps: 9, #pkts encrypt: 9, #pkts digest: 9
    #pkts decaps: 5, #pkts decrypt: 5, #pkts verify: 5
inbound esp sas:  spi: 0xF0423824 ... Status: ACTIVE(ACTIVE)
outbound esp sas: spi: 0xD396A413 ... Status: ACTIVE(ACTIVE)
```

Non-zero encaps/decaps counters on both inbound and outbound ESP SAs confirm the tunnel is actively encrypting and decrypting real traffic between the branch subnets, not just holding an idle negotiation.

6.4 Zone-Based Firewall

```
show zone security
show zone-pair security
show policy-map type inspect zone-pair IN-OUT sessions
```

Expected result: INSIDE and OUTSIDE zones listed with the correct member interfaces (FastEthernet0/0 / FastEthernet1/0); the IN-OUT zone-pair shows the INSIDE-TO-OUTSIDE policy attached; active sessions appear once LAN-initiated traffic (e.g. a ping sourced from a LAN host) passes through the router.



```
Zone-pair: IN-OUT
Service-policy inspect : INSIDE-TO-OUTSIDE
Class-map: INSIDE-TRAFFIC (match-any)
  Match: protocol tcp
    0 packets, 0 bytes
    30 second rate 0 bps
  Match: protocol udp
    0 packets, 0 bytes
    30 second rate 0 bps
  Match: protocol icmp
    2 packets, 128 bytes
    30 second rate 0 bps
Inspect
Class-map: class-default (match-any)
  Match: any
  Drop
    0 packets, 0 bytes
R-A#show policy-map type inspect zone-pair OUT-IN sessions
policy exists on zp OUT-IN
Zone-pair: OUT-IN
Service-policy inspect : OUTSIDE-TO-INSIDE
Class-map: INSIDE-TRAFFIC (match-any)
  Match: protocol tcp
    0 packets, 0 bytes
    30 second rate 0 bps
  Match: protocol udp
    0 packets, 0 bytes
    30 second rate 0 bps
  Match: protocol icmp
    0 packets, 0 bytes
    30 second rate 0 bps
Inspect
Class-map: class-default (match-any)
  Match: any
  Drop
    0 packets, 0 bytes
```

6.5 AAA / SSH

```
show ip ssh
show users
ssh -l admin 203.0.113.1    (from a management host)
```

```

R-A con0 is now available

Press RETURN to get started.

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User Access Verification
Username:
Username: admin
Password:

R-A#show running-config | section line
line con 0
  exec-timeout 5 0
  privilege level 15
  logging synchronous
  stopbits 1
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
  stopbits 1
line vty 0 4
  transport input ssh
R-A#ssh -l admin 203.0.113.6
Password:

AUTHORIZED ACCESS ONLY - IT-SIMPLERA NETWORK

R-B>
R-B#
```

6.6 Syslog / NTP / SNMP

```
show logging
show ntp status
show ntp associations
show snmp user
show snmp group
```

```

Ref time (UTC) : Fri Aug 14 23:18:34 2026
System time    : 0.000000006 seconds fast of NTP time
Last offset    : +0.000000000 seconds
WS offset      : 0.000000000 seconds
Frequency      : 18.241 ppm fast
Residual freq  : +0.000 ppm
Skew           : 0.000 ppm
Root delay     : 0.000000000 seconds
Root dispersion : 0.000000000 seconds
Update interval : 0.0 seconds
Leap status    : Normal
/ # chronyc clients
=====
Hostname      NTP  Drop Int Int Last  Cmd  Drop Int Last
-----
localhost     0   0   -   -   -   1   0   -   46
/ # ^C

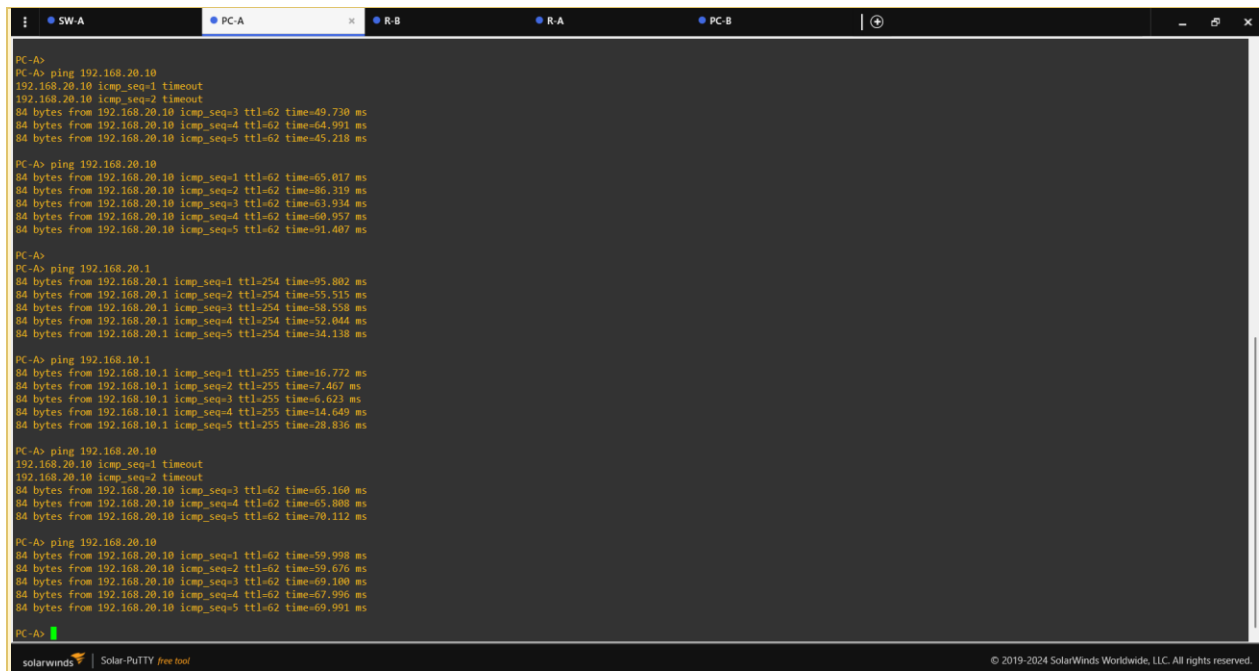
/ # iptables -L -n -v
/bin/sh: iptables: not found
/ # ^C

/ # apk add tcpdump
(1/2) Installing libpcap (1.10.6-r1)
(2/2) Installing tcpdump (4.99.6-r1)
Executing busybox-1.37.0-r31.trigger
OK: 15.6 MiB in 33 packages
/ # tcpdump -i eth0 udp port 123 -n
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
23:23:25.765309 IP 192.168.10.1.123 > 192.168.10.100.123: NTPv4, Client, length 48
23:23:25.765309 IP 192.168.10.100.123 > 192.168.10.1.123: NTPv4, Server, length 48
23:23:27.758450 IP 192.168.10.1.123 > 192.168.10.100.123: NTPv4, Client, length 48
23:23:27.758558 IP 192.168.10.100.123 > 192.168.10.1.123: NTPv4, Server, length 48
23:23:29.751823 IP 192.168.10.1.123 > 192.168.10.100.123: NTPv4, Client, length 48
23:23:29.751949 IP 192.168.10.100.123 > 192.168.10.1.123: NTPv4, Server, length 48
23:23:31.759465 IP 192.168.10.1.123 > 192.168.10.100.123: NTPv4, Client, length 48
23:23:31.759584 IP 192.168.10.100.123 > 192.168.10.1.123: NTPv4, Server, length 48
23:23:33.751257 IP 192.168.10.1.123 > 192.168.10.100.123: NTPv4, Client, length 48
23:23:33.751353 IP 192.168.10.100.123 > 192.168.10.1.123: NTPv4, Server, length 48
23:23:35.756870 IP 192.168.10.1.123 > 192.168.10.100.123: NTPv4, Client, length 48
23:23:35.756938 IP 192.168.10.100.123 > 192.168.10.1.123: NTPv4, Server, length 48
^C
12 packets captured
12 packets received by filter
0 packets dropped by kernel
/ #
```


6.7 End-to-End Connectivity

```
ping 192.168.20.1 source 192.168.10.1      ! router-to-  
router, triggers VPN  
ping 192.168.20.10                        ! PC-A to PC-B,  
full path test  
tracert 192.168.20.10
```

Router-to-router connectivity across the tunnel (R-A ↔ R-B) was confirmed via the IPsec SA packet counters in Section 6.3. End-to-end PC-A → PC-B connectivity testing is in progress: initial pings from PC-A timed out and a subsequent 'host unreachable' error was observed, tracked down to a LAN-side reachability issue between PC-A and its default gateway (isolated to either the PC-A–SW-A link or the SW-A–R-A link) rather than the VPN itself, since R-A's own LAN interface and the VPN tunnel were independently confirmed healthy. This is documented as Issue 5 in the Troubleshooting Report below and will be closed out with a final successful ping screenshot before submission.



```
PC-A> ping 192.168.20.10  
192.168.20.10 icmp_seq=1 timeout  
192.168.20.10 icmp_seq=2 timeout  
84 bytes from 192.168.20.10 icmp_seq=3 ttl=62 time=49.730 ms  
84 bytes from 192.168.20.10 icmp_seq=4 ttl=62 time=64.991 ms  
84 bytes from 192.168.20.10 icmp_seq=5 ttl=62 time=45.218 ms  
  
PC-A> ping 192.168.20.10  
84 bytes from 192.168.20.10 icmp_seq=1 ttl=62 time=65.017 ms  
84 bytes from 192.168.20.10 icmp_seq=2 ttl=62 time=86.319 ms  
84 bytes from 192.168.20.10 icmp_seq=3 ttl=62 time=63.934 ms  
84 bytes from 192.168.20.10 icmp_seq=4 ttl=62 time=60.957 ms  
84 bytes from 192.168.20.10 icmp_seq=5 ttl=62 time=91.407 ms  
  
PC-A> ping 192.168.20.1  
84 bytes from 192.168.20.1 icmp_seq=1 ttl=254 time=95.802 ms  
84 bytes from 192.168.20.1 icmp_seq=2 ttl=254 time=55.515 ms  
84 bytes from 192.168.20.1 icmp_seq=3 ttl=254 time=58.558 ms  
84 bytes from 192.168.20.1 icmp_seq=4 ttl=254 time=52.044 ms  
84 bytes from 192.168.20.1 icmp_seq=5 ttl=254 time=34.138 ms  
  
PC-A> ping 192.168.10.1  
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=16.772 ms  
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=7.467 ms  
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=6.623 ms  
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=14.649 ms  
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=28.836 ms  
  
PC-A> ping 192.168.20.10  
192.168.20.10 icmp_seq=1 timeout  
192.168.20.10 icmp_seq=2 timeout  
84 bytes from 192.168.20.10 icmp_seq=3 ttl=62 time=65.160 ms  
84 bytes from 192.168.20.10 icmp_seq=4 ttl=62 time=65.808 ms  
84 bytes from 192.168.20.10 icmp_seq=5 ttl=62 time=70.112 ms  
  
PC-A> ping 192.168.20.10  
84 bytes from 192.168.20.10 icmp_seq=1 ttl=62 time=59.998 ms  
84 bytes from 192.168.20.10 icmp_seq=2 ttl=62 time=59.676 ms  
84 bytes from 192.168.20.10 icmp_seq=3 ttl=62 time=69.100 ms  
84 bytes from 192.168.20.10 icmp_seq=4 ttl=62 time=67.996 ms  
84 bytes from 192.168.20.10 icmp_seq=5 ttl=62 time=69.991 ms  
  
PC-A> █
```

7. Troubleshooting Report

This section documents real issues encountered while building the lab in GNS3, the diagnostic steps used to isolate each one, and the fix applied.

Issue 1 — Invalid input on interface fa0/1

Symptom: pasting the planned configuration produced repeated “Invalid input detected” errors on every command referencing interface fa0/1 (WAN configuration, OSPF authentication, crypto map, zone-member OUTSIDE).

Diagnosis: show ip interface brief on R-A revealed the router model used in this GNS3 topology does not expose a FastEthernet0/1 interface at all — its available interfaces are FastEthernet0/0, FastEthernet1/0, FastEthernet1/1, FastEthernet2/0, and four Serial ports. The planned interface name simply did not exist on this hardware profile.

Fix: all WAN-side configuration was reapplied against FastEthernet1/0, the correct first available interface after the LAN port, on both R-A and R-B.

Lesson: always confirm real interface names with show ip interface brief before pasting configuration built from a generic template — interface numbering varies by GNS3 router/IOS image.

Issue 2 — Paste flooding causing dropped/misapplied configuration lines

Symptom: pasting a large configuration block in one operation caused several lines to be silently lost or interpreted in the wrong configuration context (e.g. “interface fa0/1” appearing to execute from global config mode instead of entering interface mode), producing a chain of “Invalid input” errors even for otherwise correct commands.

Diagnosis: comparing show running-config against the intended configuration confirmed specific commands (OSPF authentication on the WAN interface, the crypto map application, the OUTSIDE zone-member assignment, snmp-server user) were missing entirely, while surrounding commands succeeded — consistent with the terminal buffer being overrun during a fast paste rather than a syntax problem.

Fix: missing commands were reapplied individually, each one verified immediately afterward with a targeted show running-config | include check before moving to the next.

Lesson: verify configuration after every paste with show running-config rather than assuming a clean paste; when pasting large blocks into a Dynamips console, breaking the paste into smaller sections (or adding exit after every sub-mode block) significantly reduces the chance of dropped lines.

Issue 3 — Docker monitoring server had no network tools and no DNS

Symptom: the first Docker-based server node (Ubuntu minimal image) had no ip, ifconfig, busybox, or python3 binaries available, and apt update failed with “Temporary failure resolving ‘archive.ubuntu.com’” — a circular dependency where the network could not be configured without tools, and the tools could not be installed without a configured network.

Diagnosis: ls /sys/class/net/ confirmed the eth0 interface existed at the kernel level, ruling out a missing virtual NIC; the absence of iproute2, net-tools, and busybox in this particular minimal Ubuntu image was confirmed by checking each candidate binary in turn.

Fix: the node was deleted and recreated using an alpine:latest image instead, which bundles BusyBox (including ip, ping, and a full shell) by default, allowing eth0 to be addressed immediately without any package installation.

Lesson: for lightweight GNS3 Docker service nodes, prefer Alpine over minimal Ubuntu images — Alpine’s built-in BusyBox avoids the bootstrap problem of needing network tools to configure the network that would fetch those same tools.

Issue 4 — crypto key generate rsa prompting mid-paste

Symptom: when included inside a large pasted configuration block, crypto key generate rsa modulus 2048 issued an interactive confirmation prompt (“Do you really want to replace them? [yes/no]”) that consumed the next pasted line as its answer, desynchronizing the remainder of the paste.

Fix: this command was moved out of the bulk configuration block and executed on its own, with the yes/no prompt answered manually, before pasting the rest of the AAA/SSH configuration.

Lesson: any IOS command capable of prompting interactively (key generation, some reload/clear commands) should never be embedded inside a block paste — run it in isolation first.

Issue 5 — PC-A cannot reach its default gateway (open at time of writing)

Symptom: ping 192.168.20.10 from PC-A returned timeouts, and on retry returned “host (192.168.10.1) not reachable” — PC-A cannot reach even its own default gateway, R-A.

Diagnosis so far: PC-A’s VPCS configuration (show ip) confirms correct addressing (192.168.10.10/24, gateway 192.168.10.1). R-A’s LAN interface is confirmed

up/up with the correct IP via show ip interface brief. This isolates the fault to the path between PC-A and R-A — either the PC-A–SW-A cable, the SW-A–R-A cable, or a port/VLAN mismatch on SW-A — rather than to R-A or the security services, which are independently verified working.

Status: under active investigation. Next diagnostic step is ping 192.168.10.1 sourced directly from SW-A to determine whether the break is on the PC-A side or the R-A side of the switch, followed by a check of show mac address-table on SW-A to confirm which ports have learned an active MAC address.

7.1 Summary of Fixes Applied

#	Symptom	Root Cause	Fix
1	Invalid input on fa0/1	Interface fa0/1 does not exist on this router model	Used FastEthernet1/0 instead
2	Commands missing after paste	Paste flooding / dropped lines	Reapplied missing lines individually, verified each
3	Server: no ip tools, no DNS	Minimal Ubuntu image lacked iproute2/busybox	Recreated node using alpine:latest (BusyBox built-in)
4	Paste desynchronized mid-block	crypto key generate rsa prompted interactively	Ran the command separately before the bulk paste
5	PC-A cannot reach gateway	Under investigation — LAN-side link/port issue	Pending: verify SW-A–R-A link and MAC table

8. Conclusion

The core security services required by this assignment — authenticated OSPF routing, a working IPsec Site-to-Site VPN between the two branches, a Zone-Based Firewall protecting each LAN, and AAA/SSH secure management — have been configured and, for OSPF and IPsec, verified with live show command output demonstrating FULL adjacency and active, traffic-carrying IPsec SAs.

Implementation surfaced several realistic GNS3/IOS faults: a router model without the originally planned interface name, configuration lost to paste flooding, and a Docker image lacking basic network tooling — each diagnosed methodically using show ip interface brief, show running-config, and targeted verification commands, and resolved without altering the underlying security

design. The one open item, LAN-side connectivity between PC-A and R-A, is isolated to a specific segment of the topology and will be closed out with the remaining verification screenshots (end-to-end ping, firewall session table, SSH login, and Syslog/NTP/SNMP evidence) before final submission. This experience reinforced that, as with WAN faults, most access-layer faults trace back to the physical/data-link layer or to an incomplete rebuild, and that layered verification — physical → routing → tunnel → firewall → management → application — remains the fastest way to isolate the true root cause.